Task #1

A horse named Epona pulls a 400-kg carriage with a force of 890 N. If there is 115 N of friction, what is the acceleration? Ignore air resistance.

Task #2

A 1300-kg car accelerates at 2.5 m/s^2. If there is 239 N of friction and 623 N of air drag, what is the force applied by the engine?

Task #3

A bucket is being lifted out of a well. It has a mass of 3.2 kg. What force is needed to make the bucket accelerate at a rate of 1.35 m/s^2?

Task #4

The Mercury-Redstone Rocket that launched Alan Shepherd, the first American Astronaut in space, had a fully-loaded mass of 29,900 kg. If the upward thrust applied by the rocket at a certain point in time was 335,000 N, what was the acceleration of the rocket?